

Flaps (F1, F2)

| Concept Summary

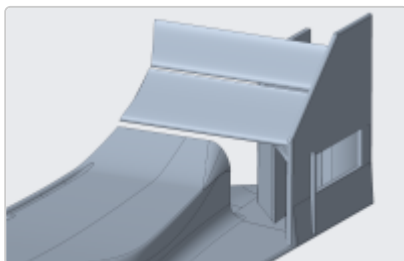
1. KIT23 Concept Evaluation

- Two flaps supporting the main wing with low pressure zone behind it
- Work like an extension of the flap while creating a new boundary layer after the main wing
- Dictate the upstream to the rear wing
- Flap geometry does not have steady curvature
- „Dead“ zone above the cooler housing → wasted downforce



2. Improvements

- Airfoil geometry changed to steady curvature and smaller angle → improved low-pressure zone of the wing, resulted in FW downforce gain (24E_UT_IT250)
- More air intake through rounding the corner of the main wing (24E_UT_IT251)



- More air intake through bigger slot gaps between the flaps

3. Unsuccessful Concepts

- Using one big flap instead of two (Schinkenkonzept)
- Pressure plate with different airfoil geometries at each side → idea: use the „dead“ zone above the cooler housing (24E_UT_IT177 until 24E_UT_IT237)



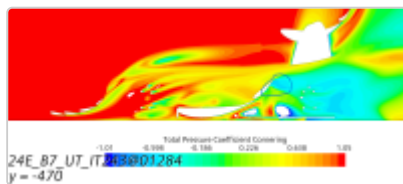
- Twisting the flap to use the „dead“ zone above the cooler housing (24E_UT_IT241 until 24E_UT_IT252)



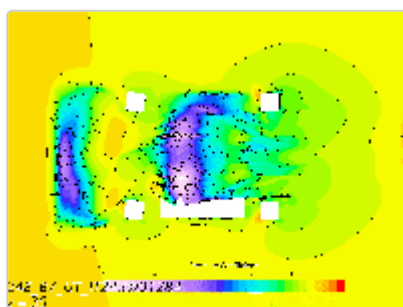
- Intake flap between the cooler housing and the main wing (24E_UT_IT268 until 24E_UT_IT271)

Why did they not work out?

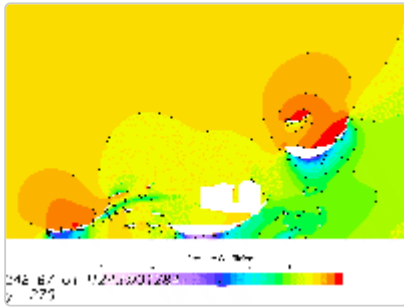
Schinkenkonzept failed because the air separates too early from the wing:



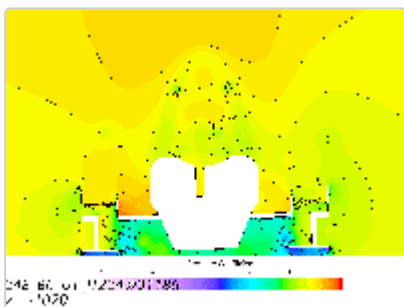
Therefore, the low-pressure zone is smaller than with two flaps. As a result, the higher counter pressure behind the main wing leads to an overall loss of downforce at the UT:



With earlier separation the main wing produced less upstream towards the RW. This led to downforce gain at the RW:

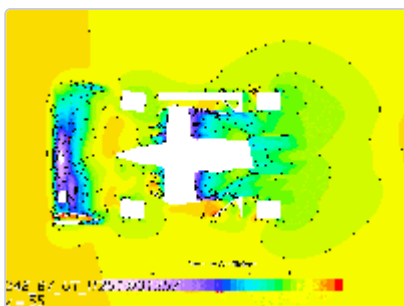


The pressure plate concept and twisted flaps had the purpose to use the „dead“ zone above the cooler housing to produce downforce. This intention worked as expected, but both concepts had the same problem. The takeaway of a flap element in the area between the main wing and the cooler housing consequently took away the low-pressure zone, too:



This zone is critical for the big low-pressure zone behind the main wing. The same problems explained above emerged. Overall, the small gain above the cooler housing did not nearly match the loss of downforce at the main wing.

The intakeflap-concept was the counter idea to maximise the low-pressure area between the main wing and the cooler housing. It led to better attachment of the air on the flaps, resulting in an extension of the low-pressure zone behind the main wing. Unfortunately, it destroyed the footplate vortex. This led to an overall small loss in terms of c_l .



4. Conclusion

The old concept of two straight flaps behind the main wing works best with the overall side wing concept of the undertray. A change of angle and shape of the airfoils paired with the main wing rounding helped the whole system to produce more downforce. This change builds on the flattening and extending of the main wing. Because of less upstreaming air, the RW benefits.